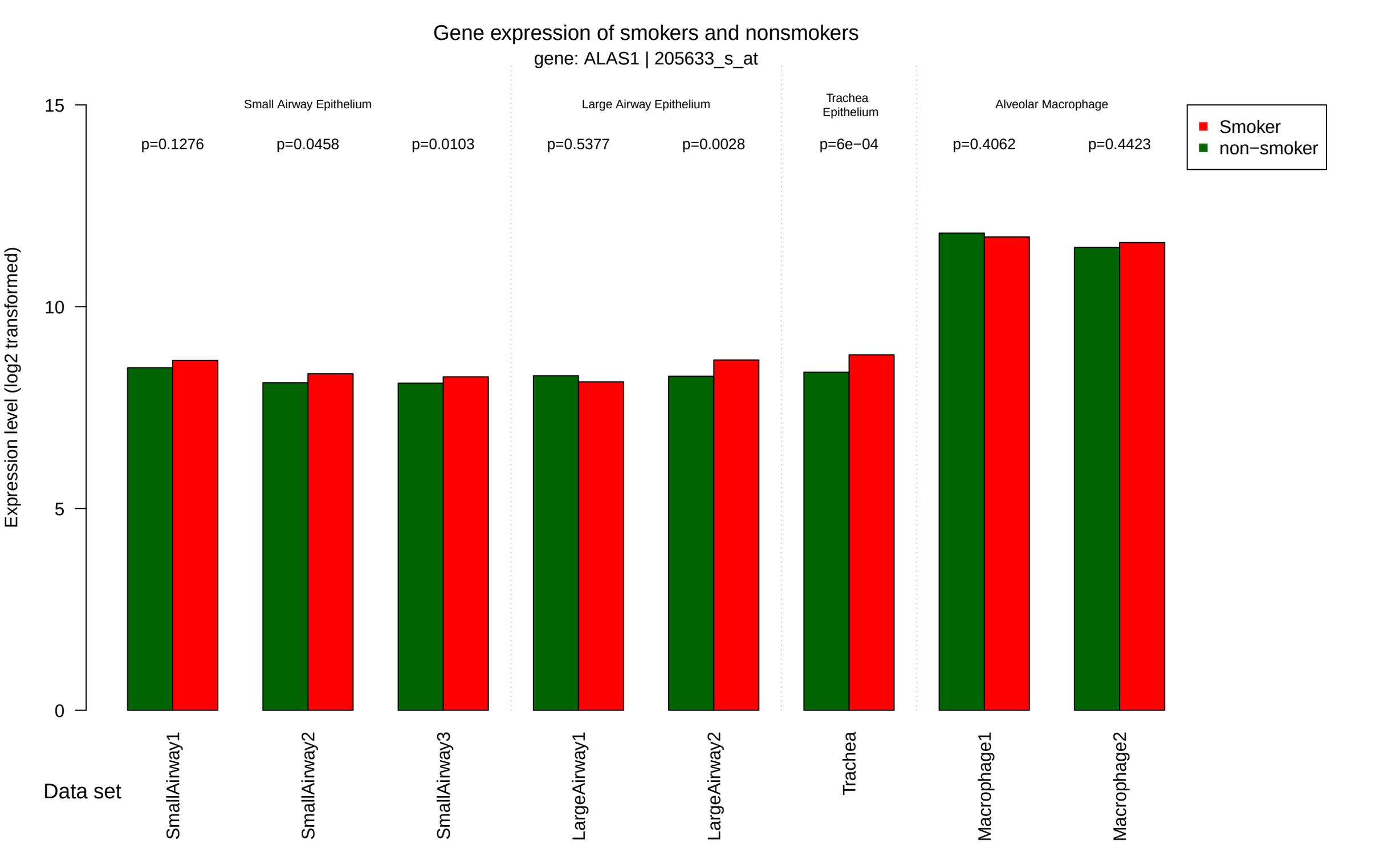
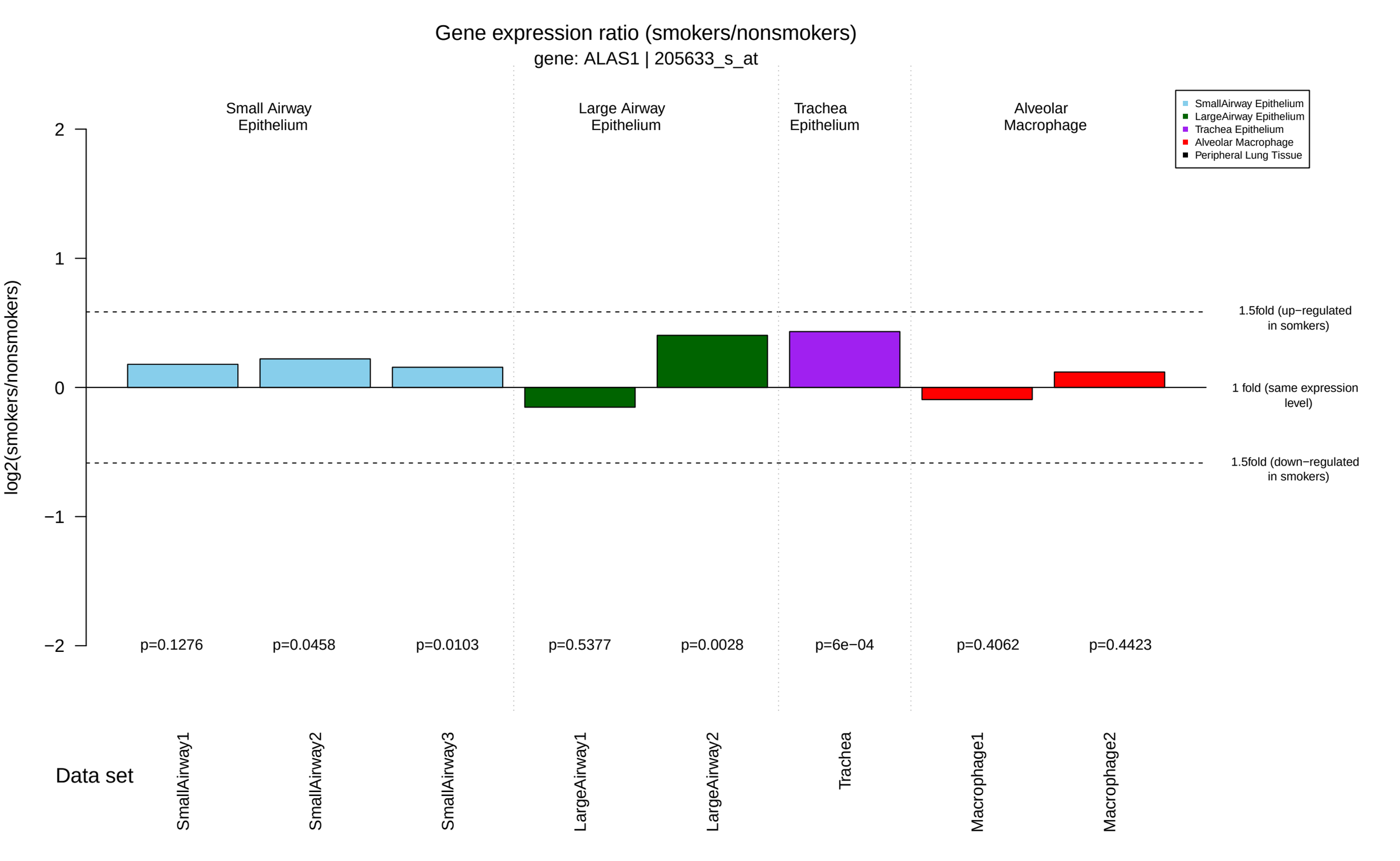
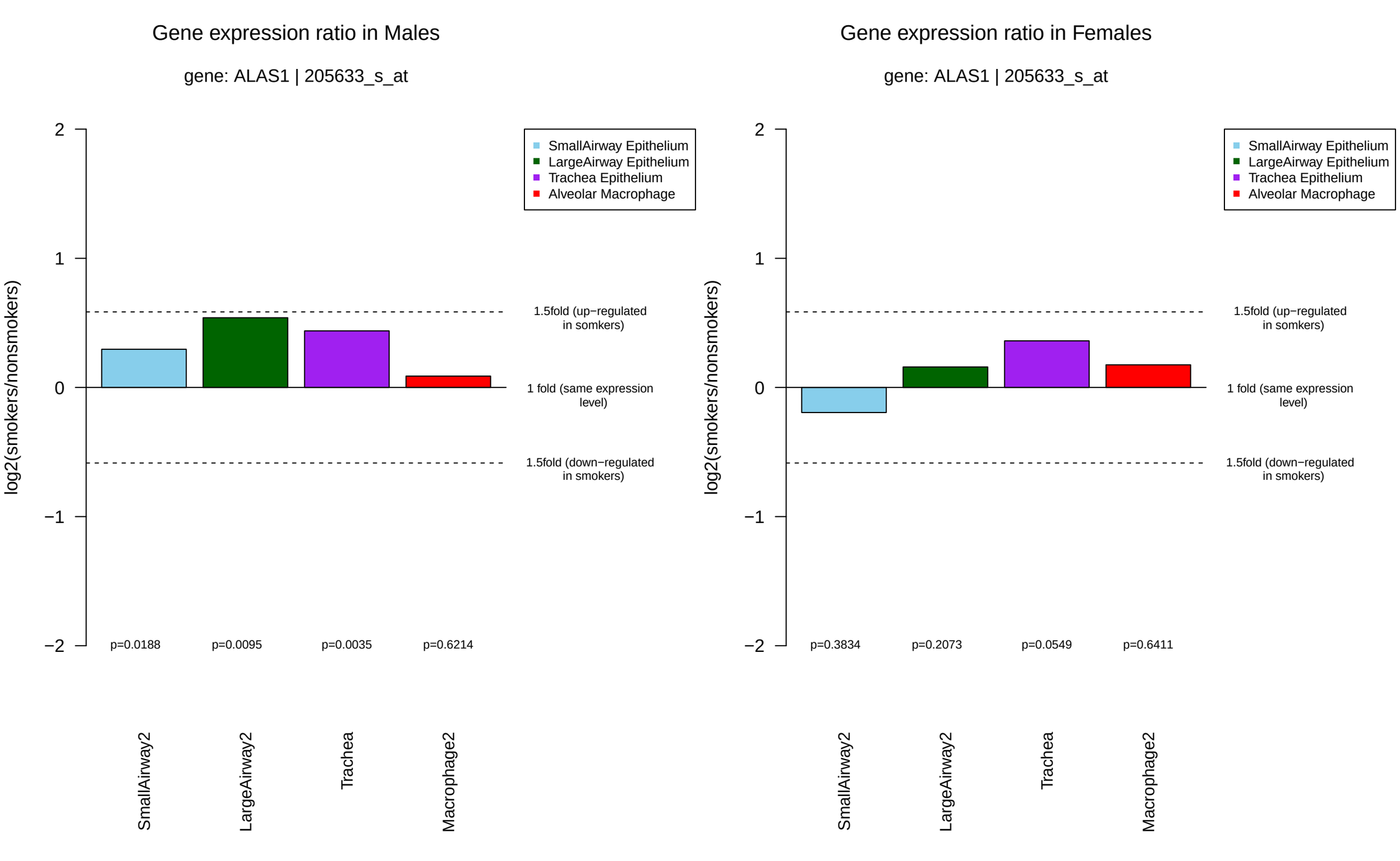
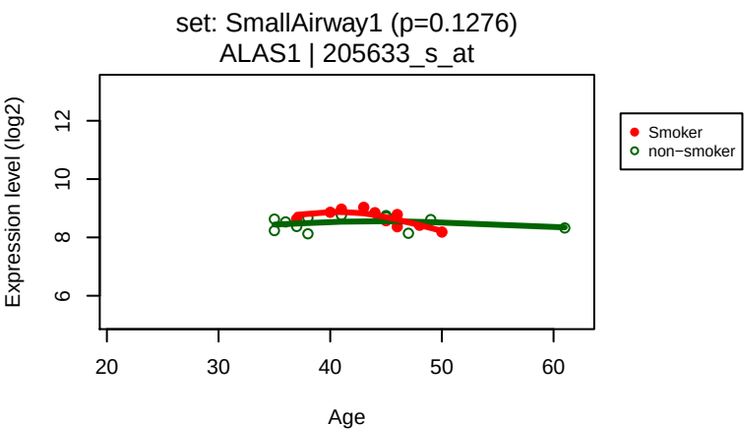
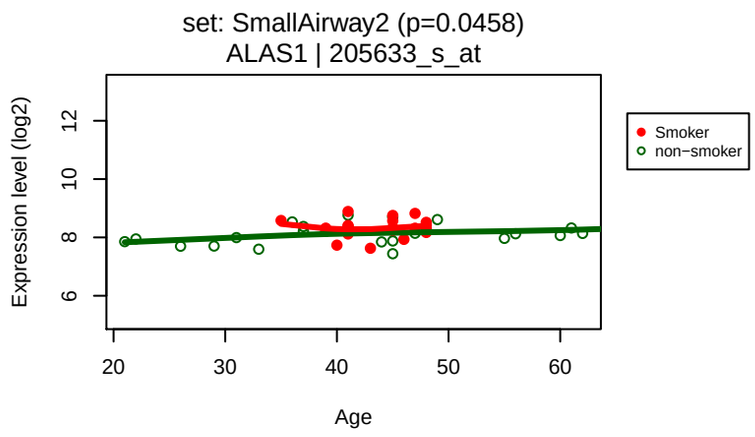
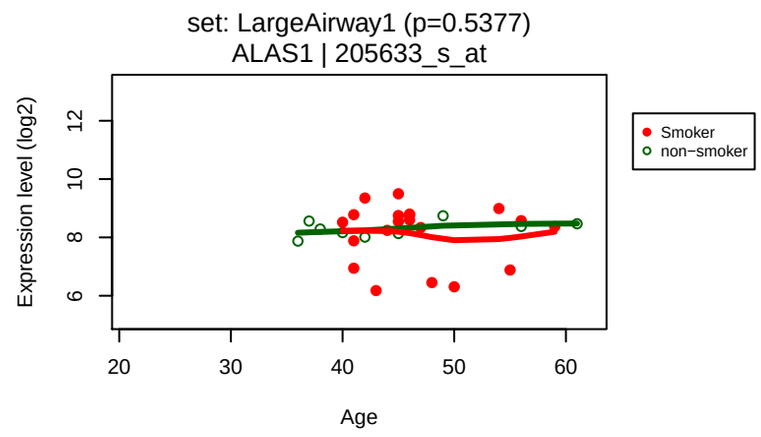

Gene expression of smokers and nonsmokers
gene: ALAS1 | 205633_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1276
p=0.0458
p=0.0103
p=0.5377
p=0.0028
p=6e−04
p=0.4062
p=0.4423
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

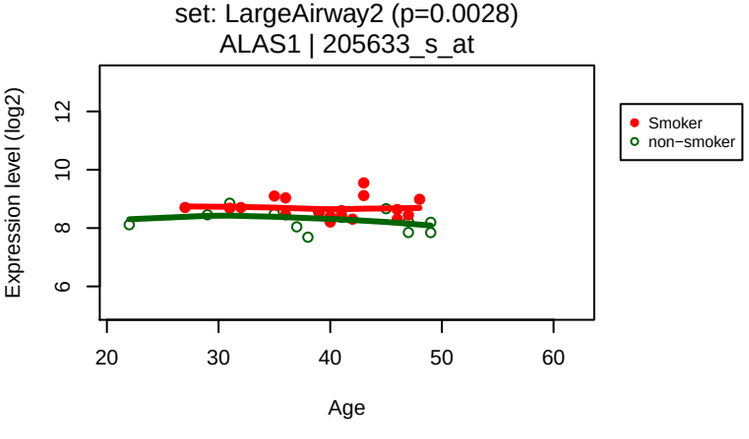

Gene expression ratio (smokers/nonsmokers)
gene: ALAS1 | 205633_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1276
p=0.0458
p=0.0103
p=0.5377
p=0.0028
p=6e−04
p=0.4062
p=0.4423
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

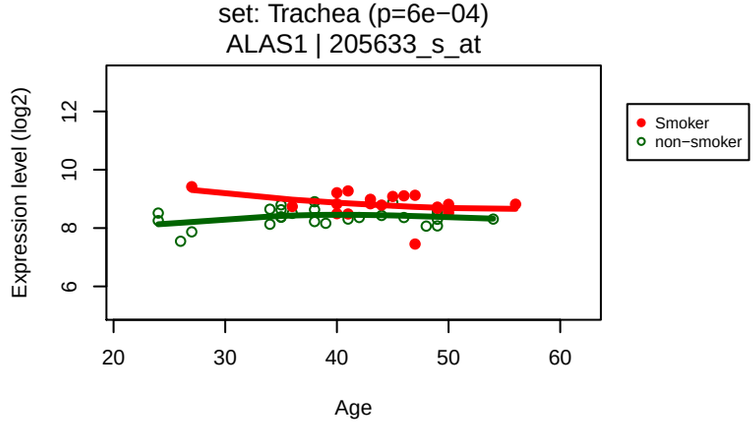

Gene expression ratio in Males
gene: ALAS1 | 205633_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0188
p=0.0095
p=0.0035
p=0.6214
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ALAS1 | 205633_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3834
p=0.2073
p=0.0549
p=0.6411
SmallAirway2
LargeAirway2
Trachea
Macrophage2

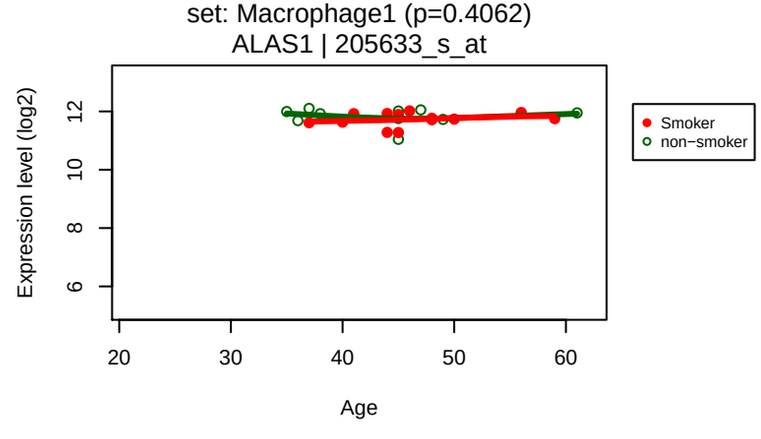

set: SmallAirway1 (p=0.1276)
ALAS1 | 205633_s_at
Expression level (log2)
Age
Smoker
non−smoker

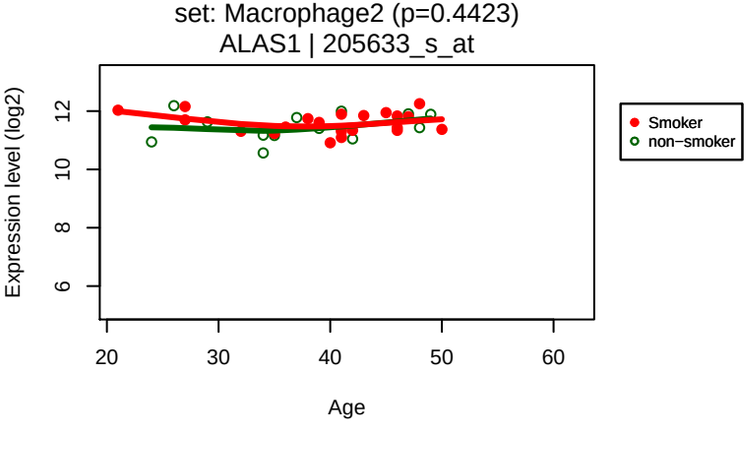

set: SmallAirway2 (p=0.0458)
ALAS1 | 205633_s_at
Expression level (log2)
Age
Smoker
non−smoker

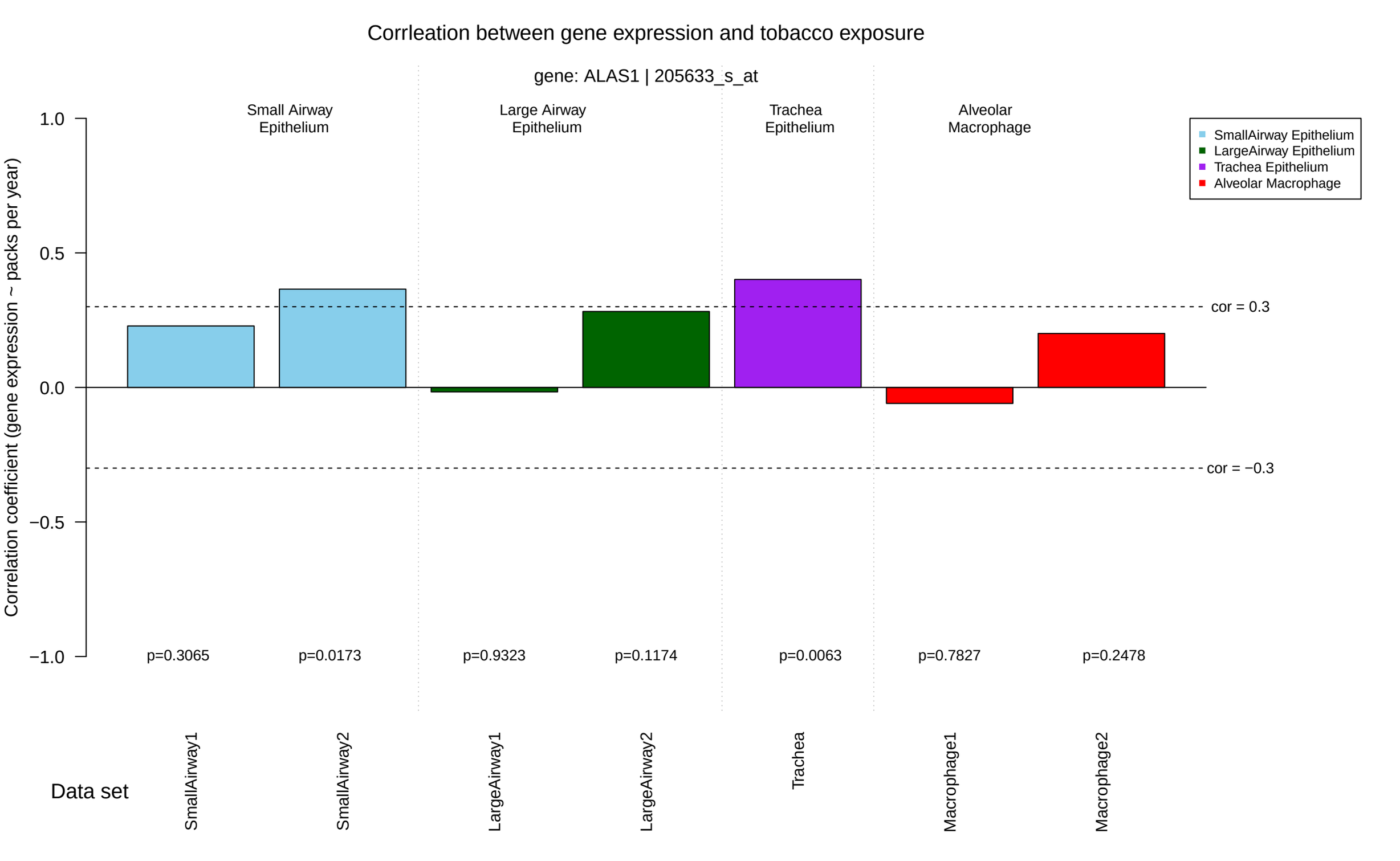

set: LargeAirway1 (p=0.5377)
ALAS1 | 205633_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0028)
ALAS1 | 205633_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=6e−04)
ALAS1 | 205633_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4062)
ALAS1 | 205633_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4423)
ALAS1 | 205633_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ALAS1 | 205633_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3065
p=0.0173
p=0.9323
p=0.1174
p=0.0063
p=0.7827
p=0.2478
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2